SEVEN SEAS BEAUTY PRODUCTS PVT LTD (HR) · MATERIAL FLOW

Architecture and Tally Integration

How the Material Flow application is built, exactly what it reads from TallyPrime, the request it sends for each piece of data, and the answer Tally gives — with files you can run yourself to show it.

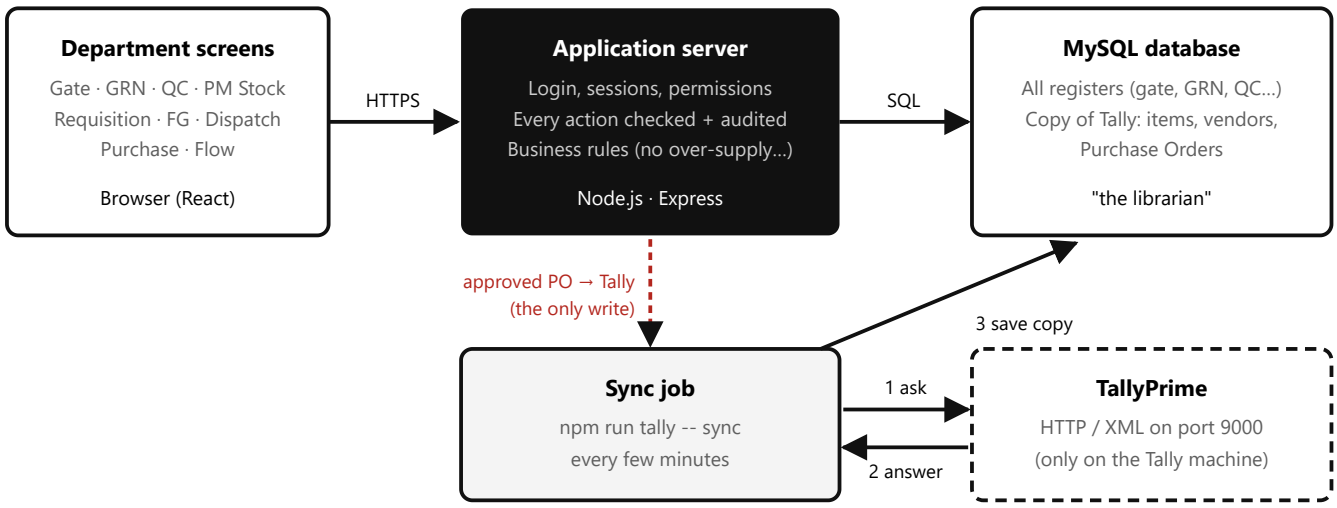
Document date	22 October 2026
Application	Seven Seas Material Flow (SOP/SCM/001 v2.1)
Tally	TallyPrime, HTTP/XML interface on port 9000 (local test copy of the company)
Runnable queries	`docs/tally-queries/` in the project — files 01–06 and `run.ps1`
Outputs shown	Real answers from the local Tally copy, captured on 22-Oct-2026, shortened to one record each

Contents

1. Architecture at a glance
2. How data comes from Tally (the sync)
3. What is read from Tally — summary
4. How to run the queries yourself
5. Query 1 — Stock items
6. Query 2 — Vendors (Sundry Creditors)
7. Query 3 — All Purchase Orders
8. Query 4 — Only Purchase Orders changed since the last sync
9. Query 5 — One Purchase Order by number
10. Request 6 — Creating an approved PO in Tally (the only write)
11. Checking the copy in MySQL
12. Safety rules

1. Architecture at a glance

The application has three parts: the department screens in the browser, the application server, and a MySQL database. TallyPrime stays where it is; a **sync job** reads it over Tally's own HTTP/XML interface and keeps a copy in MySQL.



The rules the design follows

- **No screen ever talks to Tally.** Screens read MySQL only, so the gate, store and QC keep working when Tally is closed, slow or offline.
- **MySQL is the librarian, Tally is the library.** The sync job asks Tally for what changed and files a copy in MySQL. The gate searches that copy.
- **Reads are Tally "Export" requests.** An Export cannot change anything in Tally; it only returns data.

- **Only what changed is read.** Every Tally record carries an `AlterID` that Tally raises on each change. The sync remembers the highest one it has seen and next time asks only for newer records.
- **One write, switched off by default.** An approved PO can be created in Tally (section 10). It only runs when `TALLY_ALLOW_WRITE=true`, which is on for the local test copy only.

2. How data comes from Tally (the sync)

The command `npm run tally -- sync` does this, in one database transaction. It is meant to run every 10 minutes from Windows Task Scheduler (to be set up on the Tally PC; today it is run by hand):

1. Reads the last AlterIDs it saw from MySQL (`app_meta`: `tally_alter_items`, `tally_alter_vendors`, `tally_alter_pos`).
2. Sends **Query 1** (stock items), **Query 2** (vendors) and **Query 4** (Purchase Orders changed since) to `http://localhost:9000`.
3. Reads each XML answer and checks it. A PO that cannot be read exactly — for example a PO number used twice — is **skipped and listed** on the Purchase screen for Accounts to fix in Tally; it is never guessed.
4. Saves the copy: items → `tally_items` and `components`; vendors → `suppliers`; POs → `tally_purchase_orders` + `tally_po_lines`.
5. Stores the new highest AlterIDs and a status record (time, counts, skipped POs) shown to Purchase and Admin.

`npm run tally -- sync --full` reads everything again (Query 1, 2 and 3) and marks POs that were deleted in Tally as cancelled — never deleted, because gate entries point to them.

3. What is read from Tally — summary

DATA	TALLY OBJECT & FILTER	FIELDS ASKED FOR	KEPT IN MYSQL	COUNT (22-OCT)
Stock items	`StockItem`, all	Name, Parent (group), BaseUnits, MasterID, AlterID, aliases	`tally_items`, `components`	4,849
Vendors	`Ledger` under Sundry Creditors	Name, Parent, MasterID, AlterID	`suppliers` (from_tally = 1)	320
Purchase Orders	`Voucher`, type "Purchase Order", FY 2025-26 to 2026-27	Date, number, party, cancelled, GUID, AlterID, all item lines	`tally_purchase_orders`, `tally_po_lines`	541 vouchers; 29 expense POs without stock items left out

Not read: sales, payments, bank, GST returns, payroll or any accounting entries. Only the three things the material flow needs.

How an item gets its code

An item's code in the app is its Tally alias when that alias is a PMS/PMR code (e.g. `PMS0842`); otherwise `T` + Tally's MasterID (e.g. `T4965`). Because the MasterID never changes, an item renamed in Tally keeps its code in the app.

4. How to run the queries yourself

Each request is a small XML file in `docs/tally-queries/`. It is POSTed to Tally's port; Tally answers with XML. Requirements: TallyPrime open on this PC with the company loaded, and its HTTP server on (F1 → Settings → Connectivity → TallyPrime acts as Server, port 9000).

FILE	WHAT IT ASKS	CHANGES TALLY?
`01-stock-items.xml`	All stock items	No — Export
`02-vendors.xml`	All vendor ledgers (Sundry Creditors)	No — Export
`03-purchase-orders.xml`	All Purchase Orders, 1-Apr-2025 to 31-Mar-2027	No — Export
`04-purchase-orders-changed-since.xml`	Only POs with AlterID above 12000	No — Export
`05-one-purchase-order.xml`	One PO by number (26SEV1244)	No — Export
`06-create-purchase-order-WRITES.xml`	Creates PO 26SEV1261	**Yes — test copy only**

Option A — PowerShell (prints a record count and saves the answer)

```
cd C:\Users\rudra\Downloads\star\star
powershell -ExecutionPolicy Bypass -File docs\tally-queries\run.ps1 docs\tally-queries\02-vendors.xml
```

```
LEDGER: 320 record(s)
Answer saved to docs\tally-queries\02-vendors.response.xml (136 KB, 337 ms)
```

Option B — curl (built into Windows 10/11)

```
curl.exe -X POST http://localhost:9000 -H "Content-Type: text/xml" --data-binary "@docs\tally-queries\02-vendor
s.xml" -o vendors-answer.xml
```

Open the saved answer in a browser or Notepad. To change what is asked — another PO number, another AlterID — edit the line marked in each query below.

5. Query 1 — Stock items

Purpose: the item master. Every item the gate, GRN and stock screens show comes from here, with Tally's exact name, unit and code.

REQUEST `docs/tally-queries/01-stock-items.xml`

```
<ENVELOPE>
 <HEADER><VERSION>1</VERSION><TALLYREQUEST>Export</TALLYREQUEST><TYPE>Collection</TYPE><ID>SSBItems</ID></HEADE
R>
 <BODY><DESC>
  <STATICVARIABLES>
   <SVCURRENTCOMPANY>Seven Seas Beauty Products Pvt Ltd (HR)</SVCURRENTCOMPANY>
   <SVEXPORTFORMAT>$$SysName:XML</SVEXPORTFORMAT>
  </STATICVARIABLES>
  <TDL><TDLMESSAGE>
   <COLLECTION NAME="SSBItems" ISMODIFY="No"><TYPE>StockItem</TYPE>
     <FETCH>Name, Parent, BaseUnits, MasterID, AlterID, LanguageName.*</FETCH></COLLECTION>
  </TDLMESSAGE></TDL>
 </DESC></BODY>
</ENVELOPE>
```

Read: a *collection* of all `StockItem` objects, returning only the listed fields. `LanguageName.*` carries the item's aliases (where the PMS code lives). The sync adds `<FILTER>` with `$AlterID > N` to get only changed items.

ANSWER 4,849 records, 2.4 MB, about 2.5 seconds. One record:

```
<STOCKITEM NAME="6ML EYELINER bottle" RESERVEDNAME="">
  <PARENT TYPE="String">EL-001 Eyeliner 6 Ml RM</PARENT>
  <BASEUNITS TYPE="String">Pcs</BASEUNITS>
  <ALTERID TYPE="Number"> 12130</ALTERID>
  <MASTERID TYPE="Number"> 4965</MASTERID>
  <LANGUAGENAME.LIST>
   <NAME.LIST TYPE="String">
    <NAME>6ML EYELINER bottle</NAME>
    <NAME>PMS0842</NAME>            <!-- the alias = our item code -->
   </NAME.LIST>
  </LANGUAGENAME.LIST>
</STOCKITEM>
```

STORED AS

TALLY_ITEMS.NAME	ALIAS	UNIT	GRP	MASTER_ID	ALTER_ID
6ML EYELINER bottle	PMS0842	Pcs	EL-001 Eyeliner 6 Ml RM	4965	12130

COMPONENTS.CODE	NAME	UNIT	TALLY_NAME
PMS0842	6ML EYELINER bottle	Pcs	6ML EYELINER bottle

6. Query 2 — Vendors (Sundry Creditors)

Purpose: the vendor list. Indents and POs in the app may only use these vendors, so every PO can also exist in Tally.

REQUEST `docs/tally-queries/02-vendors.xml`

```
<ENVELOPE>
 <HEADER><VERSION>1</VERSION><TALLYREQUEST>Export</TALLYREQUEST><TYPE>Collection</TYPE><ID>SSBVendors</ID></HEA
DER>
 <BODY><DESC>
  <STATICVARIABLES>
   <SVCURRENTCOMPANY>Seven Seas Beauty Products Pvt Ltd (HR)</SVCURRENTCOMPANY>
   <SVEXPORTFORMAT>$$SysName:XML</SVEXPORTFORMAT>
  </STATICVARIABLES>
  <TDL><TDLMESSAGE>
   <COLLECTION NAME="SSBVendors" ISMODIFY="No"><TYPE>Ledger</TYPE>
     <FETCH>Name, Parent, MasterID, AlterID</FETCH><FILTER>SSBVendorsFilter</FILTER></COLLECTION>
   <SYSTEM TYPE="Formulae" NAME="SSBVendorsFilter">$$IsBelongsTo:$$GroupSundryCreditors</SYSTEM>
  </TDLMESSAGE></TDL>
 </DESC></BODY>
</ENVELOPE>
```

The filter `$$IsBelongsTo:$$GroupSundryCreditors` keeps only ledgers under Sundry Creditors, including its sub-groups such as "S.C For Goods". Customers, banks and expense ledgers are not returned.

ANSWER 320 records, 136 KB, about 0.3 seconds. One record:

```
<LEDGER NAME="20 Microns Nano Minerals Limited" RESERVEDNAME="">
  <PARENT TYPE="String">S.C For Goods</PARENT>
  <ALTERID TYPE="Number"> 15312</ALTERID>
  <MASTERID TYPE="Number"> 5753</MASTERID>
  <LANGUAGENAME.LIST>
   <NAME.LIST TYPE="String"><NAME>20 Microns Nano Minerals Limited</NAME></NAME.LIST>
  </LANGUAGENAME.LIST>
</LEDGER>
```

STORED AS

SUPPLIERS.NAME	APPROVED	FROM_TALLY
20 Microns Nano Minerals Limited	1	1

7. Query 3 — All Purchase Orders

Purpose: the POs the gate receives against. Each PO comes with its item lines (item, quantity, rate, amount, due date).

REQUEST `docs/tally-queries/03-purchase-orders.xml`

```
<ENVELOPE>
 <HEADER><VERSION>1</VERSION><TALLYREQUEST>Export</TALLYREQUEST><TYPE>Collection</TYPE><ID>SSB Purchase Orders
</ID></HEADER>
 <BODY><DESC>
  <STATICVARIABLES>
   <SVCURRENTCOMPANY>Seven Seas Beauty Products Pvt Ltd (HR)</SVCURRENTCOMPANY>
   <SVFROMDATE TYPE="Date">20250401</SVFROMDATE>      <!-- period: from -->
   <SVTODATE TYPE="Date">20270331</SVTODATE>          <!-- period: to -->
   <SVEXPORTFORMAT>$$SysName:XML</SVEXPORTFORMAT>
  </STATICVARIABLES>
  <TDL><TDLMESSAGE>
   <COLLECTION NAME="SSB Purchase Orders" ISMODIFY="No">
    <TYPE>Voucher</TYPE>
    <FETCH>Date, VoucherNumber, VoucherTypeName, PartyLedgerName, IsCancelled, GUID, AlterID,
           AllInventoryEntries.*</FETCH>
    <FILTER>SSBIsPO</FILTER>
   </COLLECTION>
   <SYSTEM TYPE="Formulae" NAME="SSBIsPO">$VoucherTypeName = "Purchase Order"</SYSTEM>
  </TDLMESSAGE></TDL>
 </DESC></BODY>
</ENVELOPE>
```

ANSWER 541 Purchase Order vouchers, 14 MB, about 7 seconds. The record format is the same as in Query 5 (next pages).

> **What the sync keeps from these 541:** 29 are expense POs without stock items (services, freight) and are left out — the gate does not receive them. PO `26SEV1028` exists twice in Tally with different contents, so both copies are skipped and listed for Accounts to correct. Everything else is stored.

8. Query 4 — Only POs changed since the last sync

Purpose: what the scheduled sync actually sends. Instead of 14 MB every time, it asks only for POs whose AlterID is above the last one it saw.

REQUEST `docs/tally-queries/04-purchase-orders-changed-since.xml` — identical to Query 3 except the filter line:

```
<SYSTEM TYPE="Formulae" NAME="SSBIsPO">($VoucherTypeName = "Purchase Order") AND $AlterID > 12000</SYSTEM>
```

The sync puts the last AlterID it stored (`app_meta.tally_alter_pos`) in place of 12000. Queries 1 and 2 get the same `$AlterID > N` filter for items and vendors.

ANSWER 9 vouchers, 448 KB, about 2 seconds:

VOUCHERNUMBER	DATE	PARTY LEDGER	ALTERID	ITEM LINES
26SEV1251	19-Oct-2026	Vikash Enterprises	12018	10
26SEV1252	19-Oct-2026	Sk Industries (Delhi)	12020	9
26SEV1253	19-Oct-2026	SHASHWAT PERFUMERS	12021	4
26SEV1254	19-Oct-2026	Makeup. Com Color Cosmetics	12022	1
26SEV1255	19-Oct-2026	Selmex Industries Private Limited	12023	1
26SEV1256	20-Oct-2026	Scope Specialities Llp	12031	1
26SEV1257	20-Oct-2026	R.K. Enterprises	12035	0 (expense PO)
26SEV1258	20-Oct-2026	Chaudhary Glasspack Private Limited	12037	1
POSEV1261	01-Oct-2026	20 Microns Nano Minerals Limited	12050	1

POSEV1261 is a test PO entered by hand in the local copy — it shows up here because it is the newest change.

9. Query 5 — One Purchase Order by number

Purpose: look at one PO exactly as Tally holds it. The app also uses this before creating a PO, to make sure Tally does not already have that number.

REQUEST `docs/tally-queries/05-one-purchase-order.xml` — identical to Query 3 except the filter line (change the number to look up another PO):

```
<SYSTEM TYPE="Formulae" NAME="SSBIsPO">($VoucherTypeName = "Purchase Order") AND $VoucherNumber = "26SEV1244"</
SYSTEM>
```

ANSWER 1 voucher, 19 KB. Shortened to the fields the app reads (Tally also sends many empty fields):

```
<VOUCHER REMOTEID="69b429c3-...-00000ec2" VCHTYPE="Purchase Order" OBJVIEW="Invoice Voucher View">
  <DATE TYPE="Date">20261017</DATE>
  <GUID>69b429c3-31ff-4064-a674-9fd299810fc5-00000ec2</GUID>
  <VOUCHERTYPENAME>Purchase Order</VOUCHERTYPENAME>
  <PARTYLEDGERNAME TYPE="String">Aplay Creation</PARTYLEDGERNAME>
  <VOUCHERNUMBER>26SEV1244</VOUCHERNUMBER>
  <ISCANCELLED TYPE="Logical">No</ISCANCELLED>
  <ALTERID TYPE="Number"> 11654</ALTERID>
  <ALLINVENTORYENTRIES.LIST>                        <!-- one per item line -->
   <STOCKITEMNAME>SS-382 barcode</STOCKITEMNAME>
   <RATE>0.10/Pcs</RATE>
   <AMOUNT>-5000.00</AMOUNT>
   <ACTUALQTY> 50000 Pcs</ACTUALQTY>
   <BILLEDQTY> 50000 Pcs</BILLEDQTY>
   <BATCHALLOCATIONS.LIST>
    <GODOWNNAME>Main Location</GODOWNNAME>
    <ORDERNO>26SEV1244</ORDERNO>
    <ORDERDUEDATE JD="46311" P="25-Oct-26">25-Oct-26</ORDERDUEDATE>
   </BATCHALLOCATIONS.LIST>
  </ALLINVENTORYENTRIES.LIST>
</VOUCHER>
```

STORED AS

TALLY_PURCHASE_ORDERS.PO_NO	PO_DATE	SUPPLIER_NAME	CANCELLED	ALTER_ID
26SEV1244	2026-10-17	Aplay Creation	0	11654

TALLY_PO_LINES.PO_NO	LINE_NO	ITEM_NAME	ITEM_CODE	QTY	UNIT
26SEV1244	1	SS-382 barcode	PMS0583	50000.000	Pcs

The item code PMS0583 is found by matching the exact Tally item name against the item master from Query 1.

10. Request 6 — Creating an approved PO in Tally (the only write)

Purpose: when the owner approves a PO in the app, the app creates the same PO in Tally, in the same shape as the company's own POs. It is switched off unless `TALLY_ALLOW_WRITE=true`.

Before sending, the app checks: the PO is approved, the vendor is a Tally ledger, every item is a Tally stock item, every line has a rate, and — using Query 5 — Tally does not already have that PO number. Afterwards it reads the PO back with Query 5 so the Purchase screen shows **IN TALLY** ✓.

REQUEST — WRITES `docs/tally-queries/06-create-purchase-order-WRITES.xml` (first item line shown; the second is the same shape)

```
<ENVELOPE>
 <HEADER><VERSION>1</VERSION><TALLYREQUEST>Import</TALLYREQUEST><TYPE>Data</TYPE><ID>Vouchers</ID></HEADER>
 <BODY>
  <DESC><STATICVARIABLES></STATICVARIABLES></DESC>
  <DATA><TALLYMESSAGE>
   <VOUCHER VCHTYPE="Purchase Order" ACTION="Create" OBJVIEW="Invoice Voucher View">
     <DATE>20261001</DATE>
     <EFFECTIVEDATE>20261001</EFFECTIVEDATE>
     <VOUCHERTYPENAME>Purchase Order</VOUCHERTYPENAME>
     <VOUCHERNUMBER>26SEV1261</VOUCHERNUMBER>
     <REFERENCE>IND-0011, IND-0012</REFERENCE>         <!-- the app's indent numbers -->
     <PARTYLEDGERNAME>Aplay Creation</PARTYLEDGERNAME>
     <PERSISTEDVIEW>Invoice Voucher View</PERSISTEDVIEW>
     <NARRATION>Raised in Seven Seas Material Flow</NARRATION>
     <ALLINVENTORYENTRIES.LIST>
      <STOCKITEMNAME>NP-303 BRUSH</STOCKITEMNAME>
      <ISDEEMEDPOSITIVE>Yes</ISDEEMEDPOSITIVE>
      <RATE>5.00/Pcs</RATE>
      <AMOUNT>-25.00</AMOUNT>
      <ACTUALQTY> 5 Pcs</ACTUALQTY>
      <BILLEDQTY> 5 Pcs</BILLEDQTY>
      <BATCHALLOCATIONS.LIST>
       <GODOWNNAME>Main Location</GODOWNNAME>
       <BATCHNAME>Primary Batch</BATCHNAME>
       <ORDERNO>26SEV1261</ORDERNO>
       <ORDERDUEDATE>30-Oct-26</ORDERDUEDATE>
       <AMOUNT>-25.00</AMOUNT>
       <ACTUALQTY> 5 Pcs</ACTUALQTY>
       <BILLEDQTY> 5 Pcs</BILLEDQTY>
      </BATCHALLOCATIONS.LIST>
      <ACCOUNTINGALLOCATIONS.LIST>
       <LEDGERNAME>Purchase-Gst</LEDGERNAME>
       <ISDEEMEDPOSITIVE>Yes</ISDEEMEDPOSITIVE>
       <AMOUNT>-25.00</AMOUNT>
      </ACCOUNTINGALLOCATIONS.LIST>
     </ALLINVENTORYENTRIES.LIST>
     <!-- … second item: NP-303 INNER CAP, 5 Pcs × 5.00 … -->
     <LEDGERENTRIES.LIST>
      <LEDGERNAME>Aplay Creation</LEDGERNAME>
      <ISDEEMEDPOSITIVE>No</ISDEEMEDPOSITIVE>
      <ISPARTYLEDGER>Yes</ISPARTYLEDGER>
      <AMOUNT>50.00</AMOUNT>                          <!-- PO total -->
     </LEDGERENTRIES.LIST>
   </VOUCHER>
  </TALLYMESSAGE></DATA>
 </BODY>
</ENVELOPE>
```

ANSWER Tally replies with an import summary. When it is created:

```
<IMPORTRESULT><CREATED>1</CREATED><ALTERED>0</ALTERED><ERRORS>0</ERRORS><EXCEPTIONS>0</EXCEPTIONS>...</IMPORTRESULT>
```

When Tally refuses, the reason is in `<LINEERROR>`; the app keeps it on the PO and the Purchase screen shows **NOT SENT TO TALLY — reason** with a **Retry** button.

> **Status on the local test copy (22-Oct-2026).** The local copy runs TallyPrime in **Educational mode** (no licence). In this mode Tally takes only the 1st and 2nd of a month; other dates answer `Voucher date is missing`. On those dates every voucher *import* — including an exact copy of one of the company's own POs — comes back as `<CREATED>0</CREATED> <EXCEPTIONS>1</EXCEPTIONS>` with no reason, while entering a PO by hand works (POSEV1261, Query 4). Importing masters works. So the request has not yet been proven end to end; the next step is a **licensed test company** (a copy restored under another name on the licensed Tally). Known gap: GST lines are not yet added to the PO.

11. Checking the copy in MySQL

These queries show the same records after the sync, side by side with Tally's answers above. Run them in HeidiSQL (Laragon → Database) on the database `seven_seas_flow`.

```
-- how much of Tally is in the app
SELECT (SELECT COUNT(*) FROM tally_items)                   AS stock_items,
       (SELECT COUNT(*) FROM suppliers WHERE from_tally = 1) AS vendors,
       (SELECT COUNT(*) FROM tally_purchase_orders)          AS purchase_orders,
       (SELECT COUNT(*) FROM tally_po_lines)                 AS po_lines;

-- Query 1: one stock item, and the app item made from it
SELECT name, alias, unit, grp, master_id, alter_id FROM tally_items WHERE name = '6ML EYELINER bottle';
SELECT code, name, unit, tally_name FROM components WHERE tally_name = '6ML EYELINER bottle';

-- Query 2: one vendor
SELECT name, approved, from_tally FROM suppliers WHERE name = '20 Microns Nano Minerals Limited';

-- Query 5: one PO and its lines
SELECT po_no, po_date, supplier_name, cancelled, alter_id, synced_at FROM tally_purchase_orders WHERE po_no =
'26SEV1244';
SELECT po_no, line_no, item_name, item_code, qty, unit FROM tally_po_lines WHERE po_no = '26SEV1244';

-- where the sync stopped last time (Query 4 asks for AlterIDs above these)
SELECT k, v FROM app_meta WHERE k LIKE 'tally%';
```

RESULT on 22-Oct-2026:

STOCK_ITEMS	VENDORS	PURCHASE_ORDERS	PO_LINES
4849	320	512	845

APP_META.K	V
tally_alter_items	18744
tally_alter_vendors	18739
tally_alter_pos	12037
tally_status	{"syncedAt":"21-10-2026 12:31","full":true,"items":{"read":4849,...},"vendors":{"read":320,...},...}

POs are never deleted from the copy: a PO removed from Tally is kept and marked cancelled, because gate entries may point to it.

12. Safety rules

- **Port 9000 stays on the Tally machine.** It has no password; a Windows Firewall rule blocks it from the network. Only the sync on the same PC uses it.
- **No Tally user name or password is stored or used** by the app. The HTTP interface does not need them.
- **Reads cannot change Tally.** Queries 1–5 are Export requests.
- **The write is off by default** and is used only on a test copy until Accounts agrees.
- **The local copy is never restored back** to the company's Tally cloud.

- **Real company data is never put in source control** or in test files; the automated tests use a fake Tally with made-up data.